
CS 421 --- Objects Activity

Name	Netid

Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

Objectives

In this activity, you will:

- implement objects using two separate techniques
- implement inheritance

Part 1 --- A Counter

Here is a counter, similar to the one in the state lecture.

```
0 def mkInc(init=0):
1     ct = init
2     def inc(delta=1):
3         nonlocal ct
4         ct = ct + delta
5         return ct
6     return inc
7
8 c1 = mkInc()
```

Problem 1) We mentioned last time (or should have, anyway) that objects and closures are very similar. In what way does `mkInc` resemble an object? What is the constructor? What are methods? Is there private and public data?

Problem 2) What is missing from this story about objects?

Part 2 --- Multiple Methods

Here is a trick to introduce multiple methods.

```
0 def mkInc(init=0):
1     ct = init
2     def inc(delta=1):
3         nonlocal ct
4         ct = ct + delta
5         return ct
6     def reset(init=0):
7         nonlocal ct
8         ct = init
9         return ct
10    return (inc,reset)
11
12 (c2,r2) = mkInc()
```

Problem 3) We now have multiple methods! Would you be happy programming with an object system like this? Why or why not?

Problem 4) Add a method `dec` to this that decrements a counter.

Part 3 --- Dictionaries

Dictionaries greatly improve our quality of life.

```
0 def mkInc(init=0):
1     ct = init
2     def inc(delta=1):
3         nonlocal ct
4         ct = ct + delta
5         return ct
6     def reset(init=0):
7         nonlocal ct
8         ct = init
9         return ct
10    return { "inc": inc, "reset": reset }
```

Problem 5) How would you add dec to this version of objects?

Problem 6) Using a dictionary allows us to simulate inheritance. Here is some starter code: try to write a ``class" mkFastInc that doubles the increment each time the inc method is called. It should call the superclass methods whenever possible.

```
0 def mkFastInc(init=0):
1     superInc = mkInc(init)
2     --- your code here!
```

Problem 7) Suppose you wanted to be able to access the state of the superclass directly. What options do you have to do that?